

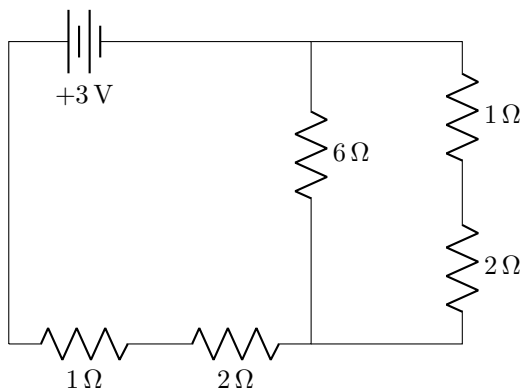
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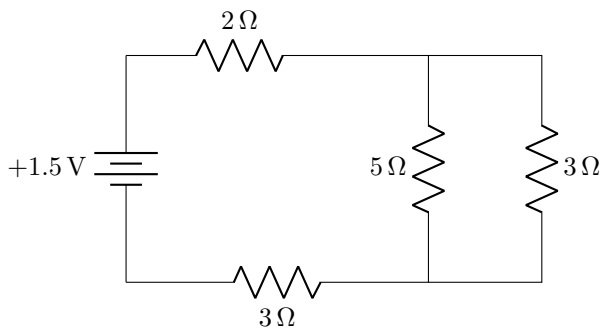
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Equivalent Resistance

1. **Consolidation:** Find the current going through the battery in the circuit below.



2. Find the current going through the battery in the circuit below.

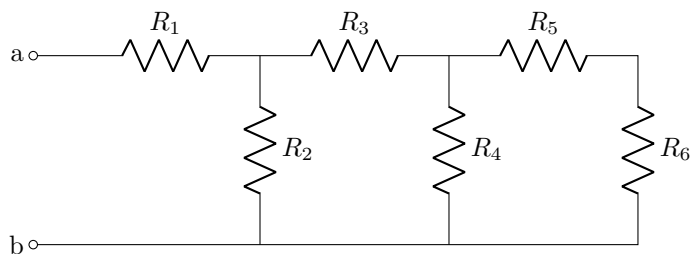


Name: _____

Date: _____

Period: _____

3. Consider the following circuit, in which $R_1 = R_3 = R_5 = R_6 = 1\text{ k}\Omega$ and $R_2 = R_4 = 2\text{ k}\Omega$.



- (a) Calculate Equivalent Resistance
(b) A 9-volt battery is connected between points a and b . Calculate the current through *each* resistor.
(c) Calculate the potential difference across each resistor.

Current I (Amperes)	Potential Difference ΔV (Volts)
Battery	
$R_1 = 1\text{ k}\Omega$	
$R_2 = 2\text{ k}\Omega$	
$R_3 = 1\text{ k}\Omega$	
$R_4 = 2\text{ k}\Omega$	
$R_5 = 1\text{ k}\Omega$	
$R_6 = 1\text{ k}\Omega$	